# PHY-112 | Principles of Physics-2

Akiful Islam (AZW)
Spring 2025 | Class #3

Department of Mathematics & Natural Sciences

BRAC UNIVERSITY
Inspiring Excellence

# Recap of the previous class!

# What we studied in Class #2

## Refer to Class #2 slides for details!



- What Electric Field Lines are and how they describe electric fields.
- How do electric charges accelerate in electric fields.
- 1D motion of charged particles in uniform electric fields.
- 2D motion of charged particles in uniform electric fields.

# The Curious Case of Electric Dipoles

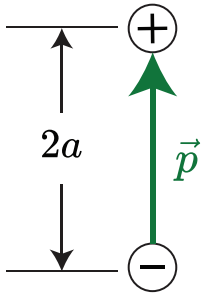
BRAC UNIVERSITY
Inspiring Excellence

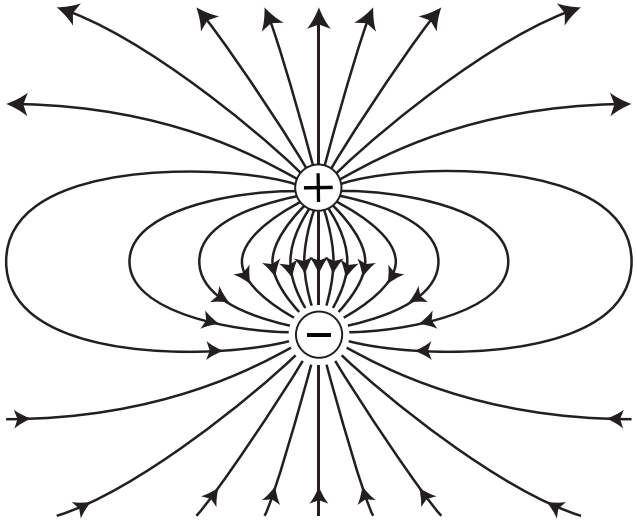
# Electric Dipole

## Two in a One system DEAL



A system consisting of two equal and opposite point charges, typically denoted as $q_+ = +ne$ and $q_- = -ne$, separated by a distance $d = 2a$.




For calculation purposes, we write $\vec{d} = (2a)\hat{d}$.

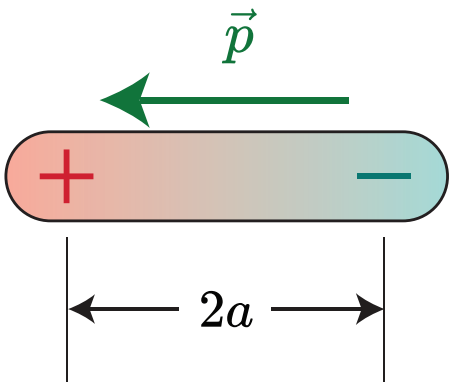
Some commonly known dipoles are water molecules, polar molecules in general, dielectric materials, capacitor plates, etc.

# Electric Dipole Moment

## Measurement of the Dipole strength



The direction of $\vec{p}$ indicates the orientation of the dipole.
The magnitude of $\vec{p}$ measures the strength/polarity of the dipole.




$$\vec{p} = q\vec{d} = q(2a)\hat{d}.$$

**Note**: $\vec{p}$ points from the $-$ve side to the $+$ve side of the dipole system.

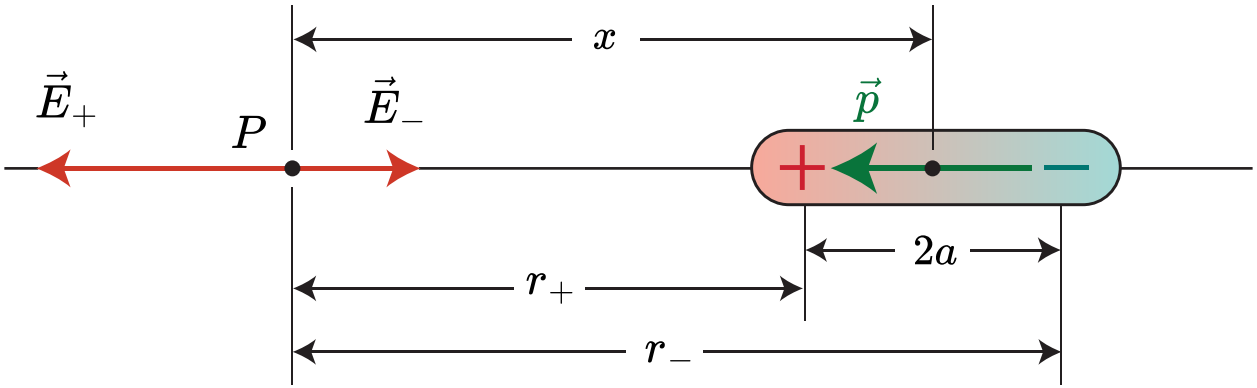
# Electric Field of an Electric Dipole

# Electric Dipole's Electric Field

## Measurement of the field strength of a dipole



An electric dipole generates an electric field that follows an *inverse cube law* (see the derivation provided with the slides).




$$\vec{E} = \frac{1}{2\pi\epsilon_0}\frac{p}{r^3}\hat{r}.$$

where $r$ is the distance (parallel to the dipole axis) from the center of the dipole to the observation point.

**Note**: Try the case when the observation is done perpendicular to the dipole axis.

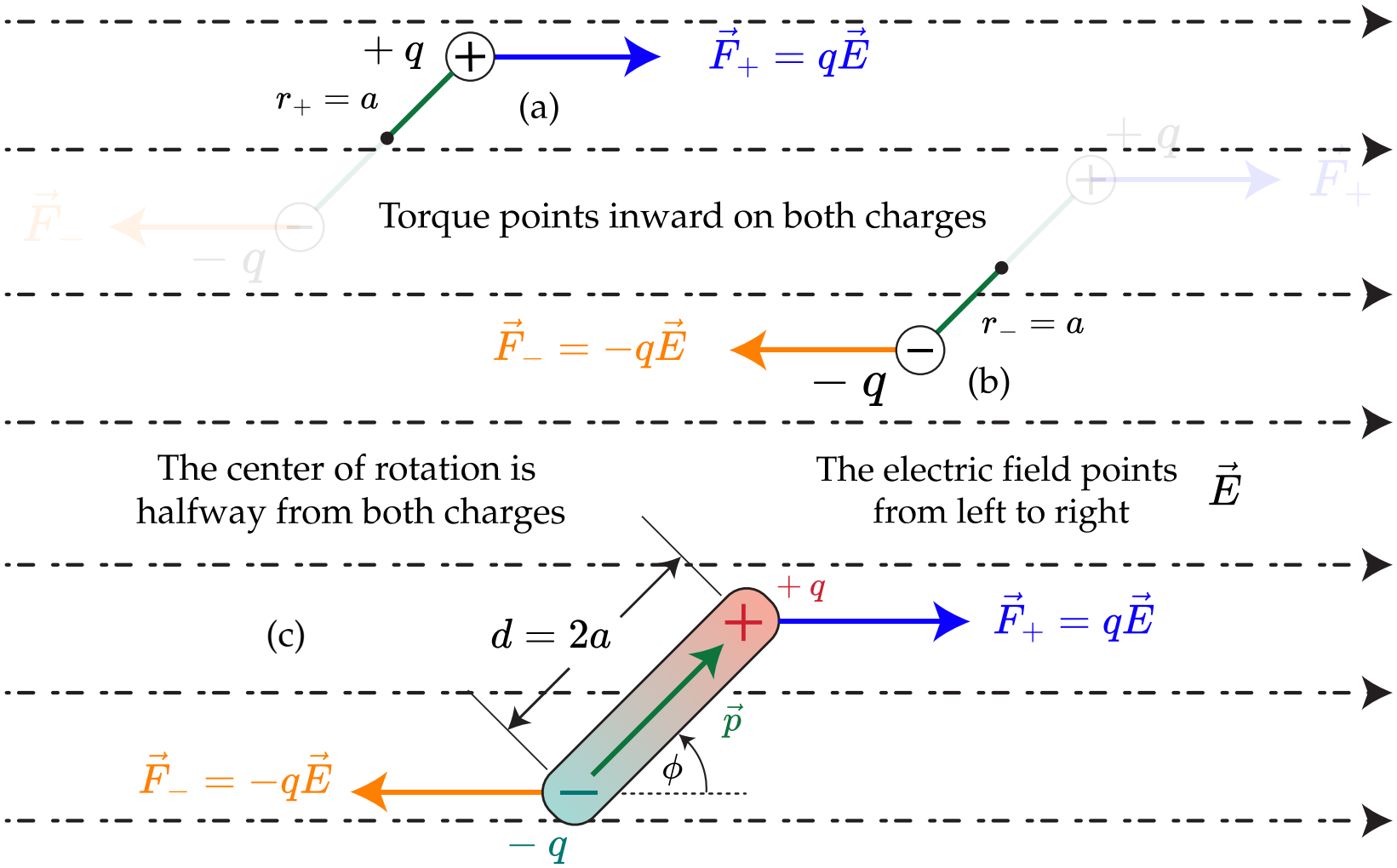
# Motion of an Electric Dipole in Electric field

BRAC UNIVERSITY
Inspiring Excellence

# Forces on Electric Dipole in a Uniform $\vec{E}$-field

## Move it!





# Forces on Electric Dipole in a Uniform $\vec{E}$-field

## Move it!



- The positive charge experiences a Coulomb force $\vec{F}_+ = +q\vec{E}$ that points along $\vec{E}$

# Forces on Electric Dipole in a Uniform $\vec{E}$-field

## Move it!



- The positive charge experiences a Coulomb force $\vec{F}_+ = +q\vec{E}$ that points along $\vec{E}$
- The negative charge of the dipole also feels an equal but opposite force $\vec{F}_- = -q\vec{E}$ that points opposite to $\vec{E}$

# Forces on Electric Dipole in a Uniform $\vec{E}$-field

## Move it!



- The positive charge experiences a Coulomb force $\vec{F}_+ = +q\vec{E}$ that points along $\vec{E}$
- The negative charge of the dipole also feels an equal but opposite force $\vec{F}_- = -q\vec{E}$ that points opposite to $\vec{E}$
- The net force on the dipole thus $\vec{F}_{\text{net}} = \vec{F}_+ + \vec{F}_- = 0$

# Forces on Electric Dipole in a Uniform $\vec{E}$-field

## Move it!



- The positive charge experiences a Coulomb force $\vec{F}_+ = +q\vec{E}$ that points along $\vec{E}$
- The negative charge of the dipole also feels an equal but opposite force $\vec{F}_- = -q\vec{E}$ that points opposite to $\vec{E}$
- The net force on the dipole thus $\vec{F}_{\text{net}} = \vec{F}_+ + \vec{F}_- = 0$
- This does not mean the dipole is motionless

# Torque on Electric Dipole in a Uniform $\vec{E}$-field

## Roll it!



- The torque for the positive charge would be
$$\vec{\tau}_+ = \vec{r}_+ \times \vec{F}_+ = a\hat{d} \times q\vec{E} = (qa)\hat{d} \times \vec{E} = \frac{1}{2}\left(\vec{p} \times \vec{E}\right)$$

# Torque on Electric Dipole in a Uniform $\vec{E}$-field

## Roll it!



- The torque for the positive charge would be
$$\vec{\tau}_+ = \vec{r}_+ \times \vec{F}_+ = a\hat{d} \times q\vec{E} = (qa)\hat{d} \times \vec{E} = \frac{1}{2}\left(\vec{p} \times \vec{E}\right)$$
- The torque for the negative charge would be
$$\vec{\tau}_- = \vec{r}_- \times \vec{F}_- = \left(-L\hat{d}\right) \times \left(-q\vec{E}\right) = (qa)\hat{d} \times \vec{E} = \frac{1}{2}\left(\vec{p} \times \vec{E}\right)$$

# Torque on Electric Dipole in a Uniform $\vec{E}$-field

## Roll it!



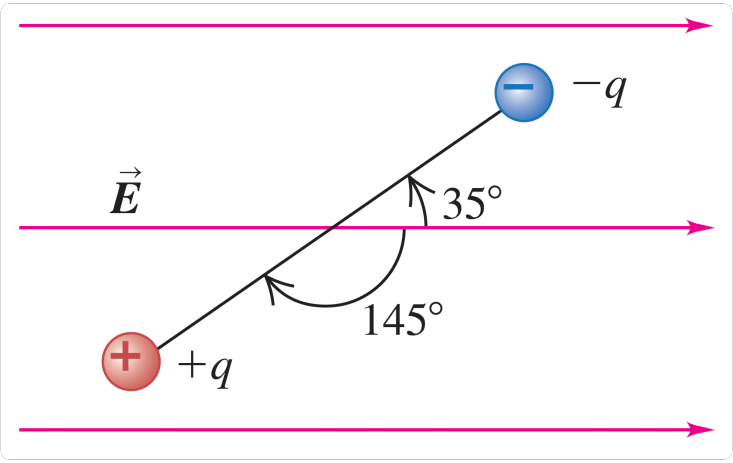
- The torque for the positive charge would be
$$\vec{\tau}_+ = \vec{r}_+ \times \vec{F}_+ = a\hat{d} \times q\vec{E} = (qa)\hat{d} \times \vec{E} = \frac{1}{2}\left(\vec{p} \times \vec{E}\right)$$
- The torque for the negative charge would be
$$\vec{\tau}_- = \vec{r}_- \times \vec{F}_- = \left(-L\hat{d}\right) \times \left(-q\vec{E}\right) = (qa)\hat{d} \times \vec{E} = \frac{1}{2}\left(\vec{p} \times \vec{E}\right)$$
- The total torque of the dipole system
$$\vec{\tau} = \vec{\tau}_+ + \vec{\tau}_- = \frac{1}{2}\left(\vec{p} \times \vec{E}\right) + \frac{1}{2}\left(\vec{p} \times \vec{E}\right) = \vec{p} \times \vec{E}$$

# Incepting Ideas (1)

## Hint: Try the extremum cases for Cross Product



Q: An electric dipole in a uniform electric field of magnitude $5.0 \times 10^5$ N C$^{-1}$ that is directed parallel to the plane of the figure. The charges are $1.6 \times 10^{-19}$ C; both lie in the plane and are separated by 0.125 nm ($0.125 \times 10^{-9}$ m). Find (a)] the net force exerted by the field on the dipole; (b) the magnitude and direction of the electric dipole moment; (c) the magnitude and direction of the torque; (d) the potential energy of the system in the position shown.



# Incepting Ideas (2)

## Hint: Try the extremum cases for Cross Product



- Q: The permanent electric dipole moment of a particular molecule is $1.1 \times 10^{-30}$ C m. What is the MAXIMUM possible torque on the molecule in a $8.0 \times 10^{8}$ N C$^{-1}$ field?

# Incepting Ideas (2)

## Hint: Try the extremum cases for Cross Product



- Q: The permanent electric dipole moment of a particular molecule is $1.1 \times 10^{-30}$ C m. What is the MAXIMUM possible torque on the molecule in a $8.0 \times 10^{8}$ N C$^{-1}$ field?
- $\longrightarrow$ Take $\phi = 90^\circ$

# Incepting Ideas (2)

## Hint: Try the extremum cases for Cross Product



- Q: The permanent electric dipole moment of a particular molecule is $1.1 \times 10^{-30}$ C m. What is the MAXIMUM possible torque on the molecule in a $8.0 \times 10^{8}$ N C$^{-1}$ field?
- $\longrightarrow$ Take $\phi = 90°$
- Q: Find the MINIMUM torque

# Incepting Ideas (2)

## Hint: Try the extremum cases for Cross Product



- Q: The permanent electric dipole moment of a particular molecule is $1.1 \times 10^{-30}$ C m. What is the MAXIMUM possible torque on the molecule in a $8.0 \times 10^{8}$ N C$^{-1}$ field?
- $\longrightarrow$ Take $\phi = 90°$
- Q: Find the MINIMUM torque
- $\longrightarrow$ Take $\phi = 0°$

# Potential Energy stored by an Electric Dipole

## Rotational work done explains this energy



$$
\begin{aligned}
U_{\text{dipole}} &= -\int \tau d\phi \\
&= -\int pE \sin\phi \; d\phi \\
&= pE\cos\phi \\
&= \vec{p}\cdot\vec{E}
\end{aligned}
$$

# Incepting Ideas (3)

## Hint: Try the extremum cases for Dot Product



- Q: The permanent electric dipole moment of a particular molecule is $1.1 \times 10^{-30}$ C m. What is the stored energy on the molecule (when placed parallel) in a $8.0 \times 10^{8}$ N $C^{-1}$?

# Incepting Ideas (3)

## Hint: Try the extremum cases for Dot Product



- Q: The permanent electric dipole moment of a particular molecule is $1.1 \times 10^{-30}$ C m. What is the stored energy on the molecule (when placed parallel) in a $8.0 \times 10^{8}$ N C$^{-1}$?
- $\longrightarrow$ Take $\phi = 0°$

# Incepting Ideas (3)

## Hint: Try the extremum cases for Dot Product



- Q: The permanent electric dipole moment of a particular molecule is $1.1 \times 10^{-30}$ C m. What is the stored energy on the molecule (when placed parallel) in a $8.0 \times 10^{8}$ N C$^{-1}$?
- $\longrightarrow$ Take $\phi = 0°$
- Q: What is the stored energy on the molecule (when placed perpendicular)

# Incepting Ideas (3)

## Hint: Try the extremum cases for Dot Product



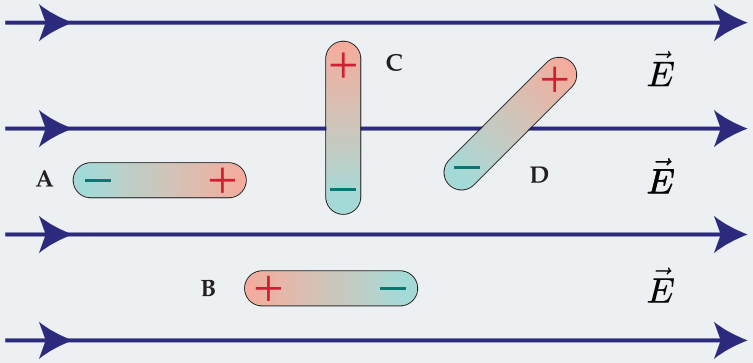
- Q: The permanent electric dipole moment of a particular molecule is $1.1 \times 10^{-30}$ C m. What is the stored energy on the molecule (when placed parallel) in a $8.0 \times 10^{8}$ N C$^{-1}$?
- $\longrightarrow$ Take $\phi = 0°$
- Q: What is the stored energy on the molecule (when placed perpendicular)
- $\longrightarrow$ Take $\phi = 90°$

# Incepting Ideas (4)

## Hint: Find $\vec{p}$ and the angle between it and $\vec{E}$



## Homework Practice Problem: Try it Yourself

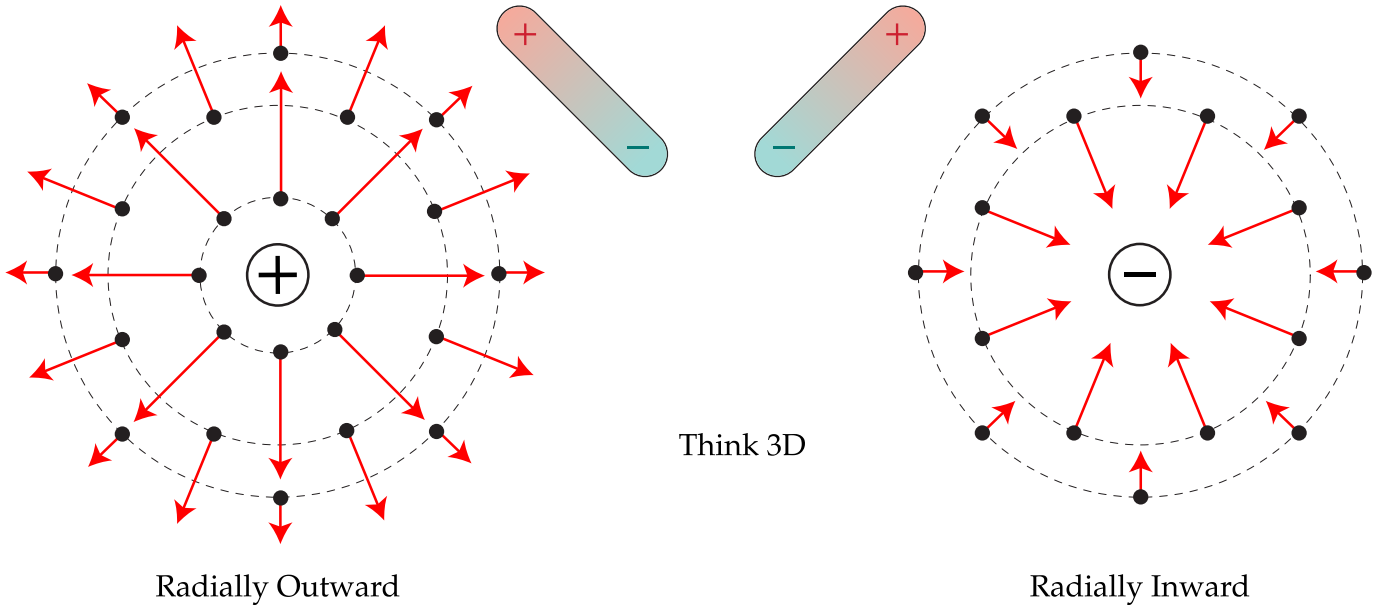



- Q: Rank the $\vec{F}_{\text{net}}$ experienced by the dipoles in descending order.
- Q: Rank the $\vec{\tau}_{\text{net}}$ experienced by the dipoles in descending order.
- Q: Rank the $U_{\text{dipole}}$ stored by the dipoles in descending order.

# Electric Dipoles in a Non-Uniform $\vec{E}$-field

## Magic? No. Physics? Yes






- Step-1: Orient $\vec{p}$ to $\vec{E}$
- Step-2: Apply Force $\vec{F}_E$ (push/pull) accordingly

# Incepting Ideas (5)

## Hint: Practice! Practice! Practice!



### Some Problems to Practice at Home on Electric Dipole

**Example Problem** 21.13, p-732, 21.14, p-733 | Young-Freedman
**Exercise Problem** 21.51, 21.55, p-740 | YF
**Checkpoint** 4, p-649 | Resnick-Halliday
**Sample Problem** 22.02, p-637, 22.05, p-650 | RH
**Exercise Problem** 56, 57, 59 p-657 | RH

# Going Continuous

BRAC UNIVERSITY
Inspiring Excellence

# One Problem Solving Strategy to Rule them All

## One strategy to bind them!



## Recipe for Discrete calculation

- Start with $\vec{E} = \left(\frac{Cq}{r^2}\right)\hat{r}$
- Superpose them: $\vec{E} = \sum_{i}^{N}\left(\frac{Cq_i}{r_i^2}\right)\hat{r}_i$ (Discrete)

## Recipe for Continuous calculation

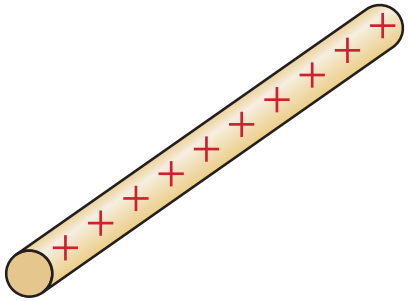
- Start with $d\vec{E} = \left(\frac{Cdq}{r_{dq}^2}\right)\hat{r}_{dq}$
- Integrate them: $\vec{E} = \int\left(\frac{Cdq}{r_{dq}^2}\right)\hat{r}_{dq}$ (Continuous)

# 3 Key $\vec{E}$ Field Sources that we may study

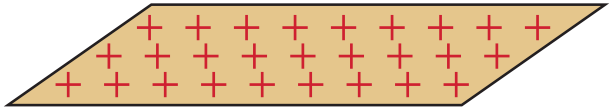
## Keep the intensity the same all over



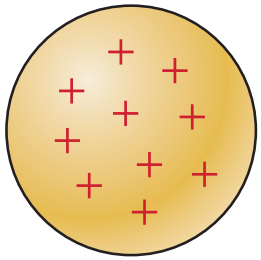
Line Charge



Surface Charge



Volume Charge



# Electric Charge Densities

## How to calculate the total charge from charge densities



### How to define the unit charge elements?

$$\text{uniform charge distribution} \equiv \begin{cases} \dfrac{dq}{dl} & \text{in one dimension (1D)} \\ \dfrac{dq}{da} & \text{in two dimension (2D)} \\ \dfrac{dq}{dV} & \text{in three dimension (3D)} \end{cases}$$

$$\text{Total Charge, } Q = \int dq = \begin{cases} \lambda dl & \text{in one dimension (1D)} \\ \sigma da & \text{in two dimension (2D)} \\ \rho dV & \text{in three dimension (3D)} \end{cases}$$

# 3 Key $\vec{E}$ Field Sources: Continuous Distributions

## Follow the ONE Rule



| Line Charge | Surface Charge | Volume Charge |
|---|---|---|
| $dq = \lambda dl$ | $dq = \sigma da$ | $dq = \rho dV$ |
| $E = C \int \frac{\lambda dl}{r^2}$ | $E = C \int \frac{\sigma da}{r^2}$ | $E = C \int \frac{\rho dV}{r^2}$ |
| $\vec{E} = \left( C \int \frac{\lambda dl}{r^2} \right) \hat{r}_{dq}$ | $\vec{E} = \left( C \int \frac{\sigma da}{r^2} \right) \hat{r}_{dq}$ | $\vec{E} = \left( C \int \frac{\rho dV}{r^2} \right) \hat{r}_{dq}$ |

# That is it for today!